



HRMS Mobile Application

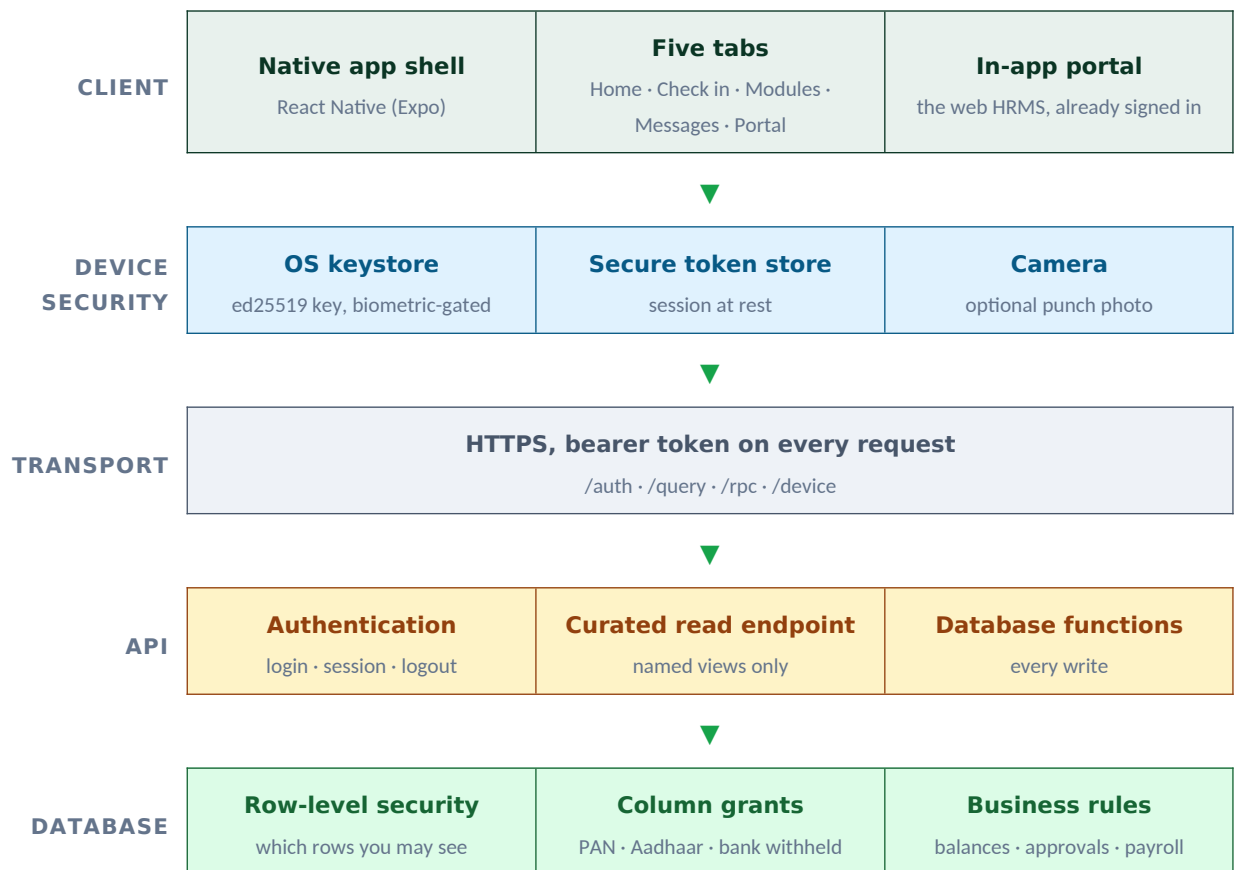
Proof of Concept · Block Diagram, Working Principle & Process Flow

Whhooohh Path LLP · com.whhooohhpath.hrms · Android & iOS · v1.0 · 7 September 2026

An HRMS that lives on the phone. Every employee signs in once, records attendance with a fingerprint or face check that the server can actually verify, and works twelve HR modules built natively for a narrow screen. Everything not worth rebuilding for a phone — payroll runs, letter generation, reports, settings — is carried inside the app as the full web portal, already signed in.

The whole system is one API and one database. There is no separate mobile backend, so a permission changed once takes effect on the phone and on the web at the same moment.

1 System Block Diagram



Reading down the stack: the client holds twelve native modules plus messaging and the embedded portal; the private key never leaves the handset; reads go through a curated endpoint that can only name known views and every write is a database function; and permission is decided in the database — which rows return, which columns are withheld, and whether the operation is allowed at all.

2 Working Principle

2.1 Attendance carries evidence, not a claim

A fingerprint check that returns a yes/no is a claim the app makes about itself, and a modified build could return yes forever. So the phone does something the server can verify instead: at enrolment a key pair is generated on the handset, the private half goes into the OS keystore behind a biometric gate, and only the public half reaches the server. Strong enough to stop a colleague punching in for an absent friend, which is the actual risk.

Order of operations at a punch	Why in that order
1. A one-time challenge is requested from the server.	So a signature cannot be captured once and replayed.
2. Reading the private key raises the phone's own biometric prompt.	The operating system enforces it. Nothing in the code can skip it.
3. The challenge is signed.	The signature is the evidence; the biometric never leaves the phone.
4. Only now is the photo taken.	A selfie of somebody who then failed the check proves nothing.
5. Signature and photo are sent and verified against the stored public key.	The server holds nothing secret and nothing biometric.

2.2 Permission is decided in the database, never in the app

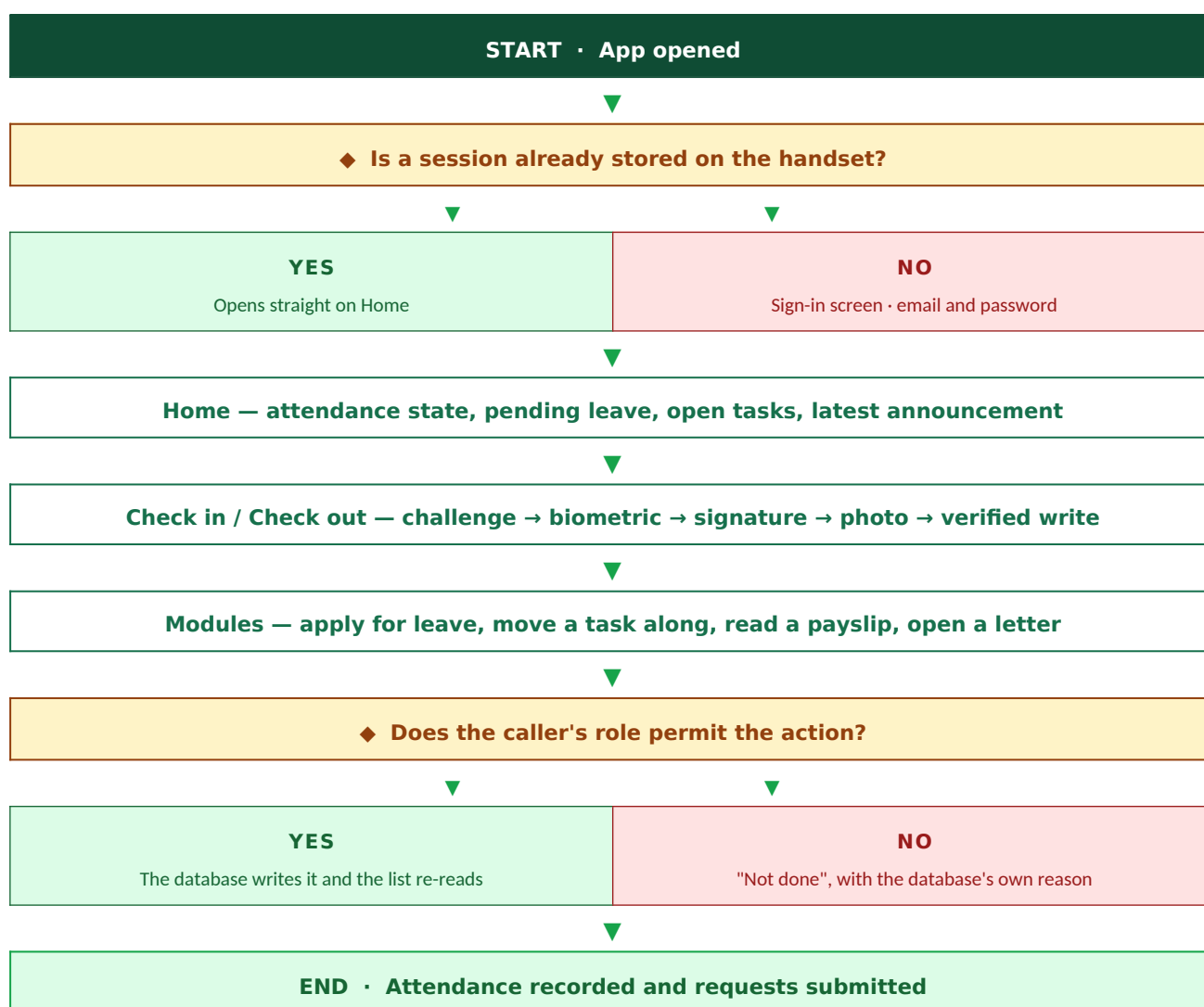
Screens hide what a role has no use for, so nobody is offered an action that will fail — but that is navigation only. The rules live in three places behind the API, and nothing bypasses them. If an employee taps Approve on their own leave request the app does not stop them; the database does, and its refusal is what appears on screen.

Mechanism	What it does
Row-level security	Decides which rows a read returns. An employee's leave list holds their own; HR's holds everyone's — same query, same screen.
Column grants	PAN, Aadhaar and bank details cannot be read from the employee table at any role. One dedicated function returns them — to HR, and to the employee themselves.
Security-definer functions	Every write and every report re-checks the caller: apply_leave checks the balance, decide_leave_request checks who may decide, each report checks for an HR role first.

2.3 Native where it earns it, portal where it does not

	What is there	Why
Native	Biometric check-in · twelve module lists · messaging	The sensor and keystore a web page cannot reach; real scrolling with no page load per tab; and an unread count that only a tab can carry.
Portal	Payroll · letters · reports · settings · recruitment · projects · company bank	Rebuilding forty screens to look at them on a phone would arrive exactly where the web app already is. The session is handed across, so it opens signed in.

3 Process Flow — Login to Recorded Action



4 Scope Covered by the Proof of Concept

Capability	Employee	HR Admin	Founder / Company Admin
Biometric attendance, own history, apply for leave	Yes	Yes	Yes
Approve or reject leave	—	Yes	Yes
Payslips	Own	Yes	Yes
Letters and certificates	Own	Issue	Issue
People, projects, tasks, assets, policies, calendar	Yes	Yes	Yes
Payroll run, salary structures, seven company reports	—	Yes	Yes
Company profile and bank account verification	—	—	Yes

The full manual — 264 pages, 72 screens, all seven roles — is the companion document to this one.